MZ 510 2018 (Quiz#2) – SOLUTION

After your help a couple of weeks ago, the fashion clothes retailer *Sarita* is now looking to expand into new markets. There are 3 candidate countries: Alphaland, Betaville, and Gammashire, but Sarita can only start operations in at most two new countries right now. Entering costs (brand marketing, regulation compliance, etc.) to Alphaland are \$10,000; to Betaville are \$13,000; and to Gammashire are \$8,000. The average margin on clothes is \$5 for a men's item, \$8 for a women's item, and \$6 for a children's item. Due to production constraints, at most 140,000 items can be manufactured and shipped, and each country requires different assortments. In Alphaland, the number of items from each department must be equal; in Betaville the number of women's clothes should be at most 60% of the total, and children's and men's items must have the same proportion. In Gammaland, no department can have more than 40% of the total number of items. Formulate an IP to decide which markets (if any) should Sarita enter, and how many items of each department will be sent to the new markets.

1. ***Decision Variables:***

x_{ij} : number of items from department i sent to country j

$i \in \{m, w, c\}$, where $m =$ men, $w =$ women, $c =$ children

$j \in \{\alpha, \beta, \gamma\}$, where $\alpha =$ Alphaland, $\beta =$ Betaville, $\gamma =$ Gammashire

y_j : 1 if Sarita enters market j; 0 otherwise

2. ***Objective Function:***

$$Z = \max_{x_{ij}, y_j} \quad \$5 \sum_j x_{m,j} + \$8 \sum_j x_{w,j} + \$6 \sum_j x_{c,j} - 10,000 y_\alpha - 13,000 y_\beta - 8,000 y_\gamma \tag{1}$$

3. ***Constraints:***

$$\sum_i \sum_j x_{ij} \leq 140,000 \tag{2}$$

$$x_{m\alpha} = x_{w\alpha} = x_{c\alpha} \tag{3}$$

$$\frac{x_{w\beta}}{\sum_i x_{i\beta}} \leq 0.6 \tag{4}$$

$$x_{m\beta} = x_{c\beta} \tag{5}$$

$$\frac{x_{m\gamma}}{\sum_i x_{i\gamma}} \leq 0.4 \tag{6}$$

$$\frac{x_{w\gamma}}{\sum_i x_{i\gamma}} \leq 0.4 \tag{7}$$

$$\frac{x_{c\gamma}}{\sum_i x_{i\gamma}} \leq 0.4 \tag{8}$$

$$\sum_i x_{ij} \leq M y_j, \quad \forall\ j \tag{9}$$

$$\sum_j y_j \leq 2 \tag{10}$$

$$y_j \text{ binary} \tag{11}$$

$$x_{ij} \geq 0, \ \forall\ i,\ j \tag{12}$$